

The effect of aerial parts of *Echinacea* on the circulating white cell levels and selected immune functions of the aging male Sprague–Dawley rat

Diana R. Cundell*, Michael A. Matrone, Paulina Ratajczak, John D. Pierce Jr.

Room 320, Search Hall, School of Science and Health, Philadelphia University, School House Lane and Henry Avenue,
Philadelphia, PA 1944, USA

Received 20 June 2002; received in revised form 11 September 2002; accepted 18 April 2003

Abstract

Echinacea is one of the most widely used over-the-counter herbal preparations that purport to “improve immune system function”, especially when taken as a short course of therapy (6–8 weeks). Since many purchasers are older individuals, a double-blind, placebo-controlled study was performed to investigate whether *Echinacea* could affect total and differential white cell counts, phagocytic activity and interleukin (IL-2) levels in 12-month-old, healthy, male Sprague–Dawley rats when administered over an 8-week period. *Echinacea* (50 mg/kg of aerial parts) mixed with peanut butter or peanut butter alone was fed to 16 rats, which were receiving regular food and water ad libitum. Cell counts and immune functions were determined on rat tail vein blood on a weekly basis. *Echinacea* significantly increased circulating total white cell counts during the first 2 weeks of administration, and IL-2 levels during the final 5 weeks of the study period ($p < 0.05$). Differential counts were altered during the entire 8-week study, with mononuclear cells significantly increased to the detriment of granulocytes ($p < 0.05$). No such changes were observed in animals given peanut butter alone. No difference was observed in phagocytic function between animals given *Echinacea* or peanut butter alone. These studies suggest that aerial components of *Echinacea* affect both mononuclear cell levels and circulating IL-2 levels in older animals.

© 2003 Elsevier Science B.V. All rights reserved.

Keywords: *Echinacea*; Rat; IL-2; White blood cells

1. Introduction

Echinacea belongs to the daisy (*Compositae*) family, and the nine North American species of these perennial wildflowers have a widespread distribution over prairies, plains and wooded areas [1]. First used

by Native Americans in the Central Plains states to treat a variety of illnesses, from respiratory and digestive problems [2] to snakebite and blood poisoning [3], the scientific properties of *Echinacea* were first studied by German scientists in the 1930s, with the result that it is now one of the leading phytomedicines used in Germany [4]. In the US, it is estimated that around 10% of the population have used it at some point to fight a cold [5], and its annual sales of an estimated US\$300 million is second only to St. John's Wort [4].

* Corresponding author. Tel.: +1-215-951-2664; fax: +1-215-951-6812.

E-mail address: CundellD@philau.edu (D.R. Cundell).

However, despite the popularity of the herb in its many forms (capsules, tables, dried root and liquid), and the fact that over 400 scientific papers have been written about *Echinacea*'s ability to "boost the immune system" both in vitro and in vivo, its true mode of action remains poorly understood. This is due, at least in part, to the diversity of methodologies and *Echinacea* preparations used. For example, evaluation of 12 human clinical trials of *Echinacea* performed between 1985 and 1989, upon which most of the claims for the herb are founded, shows that two were completely open trials and, therefore, open to both subject and experimenter bias [4]. In addition, five used stabilized juice or tinctures, whereas available over-the-counter preparations in the United States are usually dried and in tablet or capsule form [4]. Studies of in vitro immune system function have been even more diverse, representing a variety of animal models (especially rats and mice), administration methods (especially intramuscular and intraperitoneal injection) and herbal components tested (variety of single purified compounds, unspecified mixtures of components) [4–10]. Several animal and human studies have suggested that *Echinacea* stimulates neutrophil and macrophage phagocytic function [6,9,10] and the production of the cytokines interleukin-1 and tumor necrosis factor [9,10]. The final issue in evaluating the activity of *Echinacea* also comes from the fact that variations have been demonstrated in both species (*Echinacea purpurea*, *Echinacea augustifolia* and *Echinacea pallida*) and plant components (aerial components or roots) being used, as well as the extraction methods used to prepare the active ingredients [4].

Although the majority of clinical trials of the herb have to date been performed on either children or young adults in the 20–30 age group [1,4] in reality, regular purchasers and users of the herb are more likely to be older individuals. In researching the scientific literature on *Echinacea*, it was also noted that no studies currently exist which describe a double-blind investigation of the efficacy of *Echinacea* on normal, healthy, aging immune functions and white cell levels. The current study, therefore, investigated an animal model involving male, laboratory rats who were 1 year old at the beginning of the study and whose expected maximal lifespan of 2 years indicated they were an aging population. Rats have been documented to demonstrate similar aging patterns to humans and are

good models of immune function decline especially in the areas of thymic function and circulating lymphocyte and macrophage activity, with changes becoming apparent as early as 13 months of age [11–13].

Echinacea (50 mg/kg of aerial parts; equivalent to recommended human dosage) mixed with peanut butter or peanut butter alone was provided each morning to 16 male, healthy, 1-year old Sprague–Dawley rats during an 8-week period. Since rats enjoy peanut butter, this was ingested as a single bolus. During the study period and for the 3 weeks prior to the study, a number of immune functions, as well as white blood cell levels, were monitored on a weekly basis using blood obtained from the tail vein. Phagocytosis was measured using a standard technique and blood interleukin-2 (IL-2) levels were assessed using a commercial enzyme-linked immunosorbent assay (ELISA). Total and differential white cell counts were also assessed each week during this period using standardized techniques.

2. Materials and methods

2.1. Description of animal model, trial design and selection of herbal therapeutic

Sixteen healthy, male, white Sprague–Dawley rats that were 12 months of age at the onset of the study were randomly assigned to receive either the therapeutic agent or vehicle alone. Administration of *Echinacea* combined with peanut butter (active agent) or peanut butter alone (vehicle) for a period of 8 weeks was preceded and followed by a "washout period" of 3 weeks, which allowed the physiology of the animal to return to baseline values. The active agent or vehicle alone was given to the rats once a day, first thing each morning, and was immediately ingested (within 10 min) to their fondness for peanut butter. Knowledge of which rats received the active agent was only available to the study coordinators (Drs. Pierce and Cundell), although the student investigators were aware of what phase of the study the rats were in (washout or therapeutic period). *Echinacea* aerial parts (Nature's Resource, CVS Pharmacy, Philadelphia, PA, capsule form; each capsule containing *Echinacea* aerial parts 1.05 g, cichoric acid 10.5 mg combined with gelatin and water) were given to the rat test group on a weight/

volume basis, at an active ingredient level equivalent to recommended human dosages (*Echinacea* 50 mg/kg and cichoric acid 0.5 mg/kg).

Monitoring and testing of dietary supplements is not covered under FDA guidelines, but is independently assessed by a consumer council (accessible at <http://www.consumerlab.com>). *Echinacea* was, therefore, obtained from Nature's Resource, a manufacturer approved by this group, which had performed evaluations for safety from pesticide contaminants, identity, quality and purity.

2.2. Blood sampling

Immune functions were assessed in the animals on a weekly basis, at the end of each week, using samples obtained from the animals' tail vein. Blood was col-

lected using 370 µl heparin-containing Caraway tubes (Fisher Scientific, Burr Ridge, IL) and then emptied into 1.5 ml Eppendorf tubes (Fisher Scientific) containing 100 µl of heparin as anticoagulant. Approximately 1 ml of blood was drawn from each animal, of which 0.5 ml was immediately microcentrifuged for 5 min and the resultant supernatant removed and stored at -80°C for later monitoring of specific immune system function via assessment of interleukin-2 (IL-2) levels. The remainder was used for differential and total white cell counts and assessment of phagocytic activity.

2.3. Immune system studies and circulating white blood cell assessment

Phagocytosis was assessed using a standard latex bead assay [14], in which the circulating monocytes

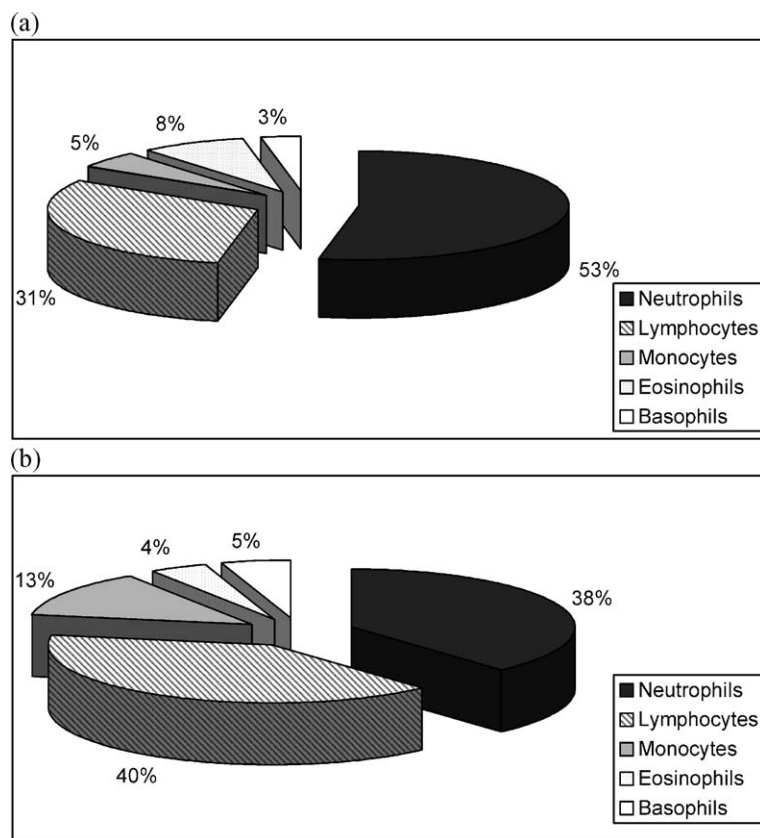


Fig. 1. Comparison of differential white cell counts during peanut butter (a) or peanut butter plus Echinacea. (b) Data shown represent mean differential white cell values obtained for 8 animals during the test period, when chow and water were provided ad libitum to all animals, who were also additionally fed each day with a single dose of either peanut butter alone or mixed with Echinacea.

and neutrophils were visually assessed as to their efficacy at ingesting latex particles. Blood samples were initially sedimented over 6% dextran (Sigma, St. Louis, MO) to separate erythrocytes from leukocytes. The supernatant containing mixed leukocytes was then microcentrifuged, resuspended in an equal volume of sterile saline, and 100 μ l of this suspension was mixed with 50 μ l of latex beads (0.1 μ m polystyrene, Sigma) and incubated for 30 min at 37 °C. Incubation was followed by three microcentrifugations and resuspensions, ensuring the removal of unphagocytosed latex beads. The percentage of leukocytes that had ingested latex beads was then determined by light microscopy.

Total and differential counts of circulating leukocytes were also performed according to standard techniques [15]. Total white blood cell count was determined using standardized hemocytometer analysis (Improved Neubauer hemocytometer, Bright Line®, American Optical, Buffalo, NY) of a Crystal Violet stained sample (Sigma). Differential white blood cells counts involved the determination of the percentages of each white blood cell under a light microscope. An air-dried blood smear was stained with Hansel Wright stain (Sigma) and washed with phosphate/sodium azide buffer (Sigma) and 95% ethanol. Once dry, cell types, including neutrophils, eosinophils, basophils, monocytes and lymphocytes, were distinguished by their appearance after staining.

Circulating IL-2 levels were measured using a commercial quantitative solid-phase sandwich immunoassay ELISA kit obtained from Anogen (Mississauga, Ontario, Canada), which had limits of detection of between 0 and 2500 pg/ml. The assay was performed in a microtiter plate whose wells were coated with a monoclonal antibody highly specific for IL-2 (no cross-reactivity with any other cytokines tested). Application of an enzyme-linked polyclonal antibody to the wells was followed by substrate and resulted in the development of color in direct proportion to the IL-2 levels of the sample or standard being assessed. The optical density of color in each well was quantitated using a Bio-Rad Microplate Reader at a 450-nm wavelength (Bio-Rad, Hercules, CA). Levels of IL-2 were determined by comparison with a standard curve of optical density versus picogram per milliliter prepared from the known IL-2 standards on each plate. All IL-2

Table 1
Effect of Echinacea administration on circulating total white cell counts

Leukocytes per milliliter		Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
P	2,472,711 $\pm$ 211,183	2,788,976 $\pm$ 269,150	2,704,854 $\pm$ 274,031	2,526,055 $\pm$ 316,605	2,843,826 $\pm$ 295,233	2,539,719 $\pm$ 216,435	2,786,701 $\pm$ 323,507	2,575,278 $\pm$ 316,477	
E	3,616,198 $\pm$ 259,015	3,425,688 $\pm$ 182,711	2,546,643 $\pm$ 484,025	2,920,338 $\pm$ 776,387	3,092,311 $\pm$ 678,727	2,934,599 $\pm$ 534,509	3,003,160 $\pm$ 286,824	2,814,306 $\pm$ 431,951	

Results shown represent the means $\pm$ S.D. of eight rats per treatment (either peanut butter, P, or peanut butter mixed with Echinacea, E), with counts being obtained from single bleeds on a weekly basis.

assays were performed on the same day to reduce experimenter error.

2.4. Statistical analyses

Data were examined using a commercially available statistical package (SPSS version 10.1 for Windows), and comparisons were made using the paired *t*-test.

3. Results

3.1. Effect of *Echinacea* on immune functions and white cell numbers

3.1.1. Circulating white cell levels

Echinacea was found to affect both total and differential circulating white cell counts during the course of the 8-week study (Fig. 1, Tables 1 and 2). Animals that were fed the peanut butter vehicle alone demonstrated mean total white cell counts of $2,654,821 \pm 255,799$, which did not vary significantly throughout the trial (Table 1). In contrast, circulating white cells in animals receiving *Echinacea* rose significantly ($p < 0.05$) during the first 2 weeks of the study compared with control (week 1 *Echinacea* and control values of $3,616,198 \pm 259,015$ and $2,472,711 \pm 211,183$ cells/ml, and week 2 values of $3,425,688 \pm 182,711$ and $2,788,976 \pm 269,150$ cells/

ml; Table 1). This increase was, however, transitory with white cell levels during the remaining 6 weeks of the trial period being similar in both groups (Table 1).

Differential white cell counts demonstrated a significant and sustained alteration ($p < 0.05$) by *Echinacea* administration throughout the 8-week period (Fig. 1, Table 2). During the 8-week study, an immediate and sustained change was observed in the distribution of circulating leukocytes with an immediate, sustained and significant increase in the percentages of lymphocytes (week 1 data for *Echinacea* and control; $39.4 \pm 7.2\%$ and $30.9 \pm 4.3\%$, respectively) and monocytes (week 1 data for *Echinacea* and control; $13.0 \pm 2.8\%$ and $4.9 \pm 1.8\%$, respectively) (Fig. 1 and Table 2). These increases were also associated with a sustained, significant and immediate decrease in the percentages of circulating neutrophils (week 1 data; $53.0 \pm 5.8\%$) and eosinophils (week 1 data for *Echinacea*; $4.0 \pm 2.7\%$), when compared with animals receiving peanut butter (week 1 data for neutrophils and eosinophils; $37.9 \pm 3.4\%$ and $8.0 \pm 2.2\%$, respectively, $p < 0.05$; Fig. 1 and Table 2). Basophil numbers in both *Echinacea* and control groups were similar during the 8-week study period (Fig. 1, Table 2).

3.1.2. Circulating interleukin-2 (IL-2) levels

Baseline levels of IL-2 measured in the animals for 3 weeks prior to administration of *Echinacea* ranged

Table 2
Effect of *Echinacea* on circulating differential white cell counts

	Percentage of each type of leukocyte observed							
	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
<i>Peanut butter</i>								
Neutrophils	53.0 ± 5.8	56.1 ± 3.1	56.9 ± 3.0	54.1 ± 3.1	51.0 ± 2.8	45.0 ± 3.3	52.3 ± 1.6	54.0 ± 2.3
Lymphocytes	30.9 ± 4.3	27.7 ± 4.4	28.3 ± 4.8	31.6 ± 7.3	32.0 ± 5.5	30.6 ± 5.7	30.9 ± 2.9	29.5 ± 4.6
Monocytes	4.9 ± 1.8	3.5 ± 2.2	3.9 ± 1.7	3.8 ± 1.3	4.8 ± 2.6	3.5 ± 1.6	4.7 ± 2.1	4.5 ± 2.3
Basophils	3.1 ± 1.8	3.4 ± 1.7	2.9 ± 2.4	3.4 ± 2.3	3.1 ± 1.6	2.9 ± 1.5	2.5 ± 1.8	3.3 ± 2.7
Eosinophils	8.0 ± 2.2	9.3 ± 1.7	7.9 ± 2.2	6.9 ± 2.1	8.0 ± 1.5	8.4 ± 1.8	8.6 ± 1.9	8.5 ± 2.2
<i>Echinacea</i>								
Neutrophils	37.9 ± 3.4	35.8 ± 3.3	33.4 ± 3.6	28.1 ± 2.0	32.9 ± 4.0	30.4 ± 5.5	31.4 ± 3.9	34.5 ± 6.1
Lymphocytes	39.4 ± 7.2	39.8 ± 4.0	43.1 ± 2.5	47.4 ± 7.5	43.4 ± 6.8	47.5 ± 10.1	46.3 ± 3.2	40.8 ± 2.0
Monocytes	13.0 ± 2.8	17.4 ± 3.8	16.4 ± 3.2	17.3 ± 2.9	14.4 ± 3.2	14.5 ± 3.7	15.8 ± 3.9	18.1 ± 3.0
Basophils	4.9 ± 3.0	2.6 ± 1.4	3.1 ± 1.1	3.3 ± 1.8	4.3 ± 2.1	4.5 ± 2.9	3.1 ± 1.2	3.4 ± 1.4
Eosinophils	4.0 ± 2.7	4.0 ± 1.7	3.3 ± 1.8	3.1 ± 1.1	4.0 ± 2.2	3.0 ± 1.2	3.1 ± 1.3	3.0 ± 1.2

Results shown represent the means $\pm$ S.D. of eight rats per treatment (either peanut butter or peanut butter mixed with *Echinacea*), with counts being obtained from single bleeds on a weekly basis.

Table 3
Effect of Echinacea on circulating IL-2 levels

	IL-2 levels (pg/ml)							
	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
Peanut butter	15 ± 21.3	8.1 ± 15.1	3.8 ± 10.7	6.2 ± 11.9	7.5 ± 13.9	8.8 ± 16.4	9.4 ± 17.8	16.3 ± 21.3
Echinacea	3.8 ± 10.7	9.4 ± 17.8	27.5 ± 27.8	1218 ± 832.5	1058 ± 771.7	517 ± 706.8	71.3 ± 81.5	40.0 ± 76.3

Results shown represent the means ± S.D. of eight rats per treatment (either peanut butter or peanut butter mixed with Echinacea), with data being obtained using a commercial ELISA.

from 0 to 35 pg/ml of blood (mean level ± S.D. of 14.4 ± 3.7 pg/ml). During the first 2 weeks of the study period, no significant effect on IL-2 levels was noted (mean values of 3.8 ± 10.7 and 9.4 ± 17.8 pg/ml, respectively; Table 3). In contrast, the remaining 6 weeks of the study produced dramatic increases in circulating IL-2 levels, which were maximal between weeks 4 and 6 (mean values of 1218 ± 832.5 , 1058 ± 771.7 and 517 ± 706.8 pg/ml, respectively; Table 3) and thereafter declined during weeks 7 and 8, although still remaining above control values (mean values of 71.3 ± 81.5 and 40.0 ± 76.3 , respectively; Table 3). Animals receiving peanut butter alone demonstrated no increase in circulating IL-2 levels, which demonstrated similar levels of >20 pg/ml during the 8-week study (Table 3).

3.1.3. Phagocytic activity in circulating leukocytes

Prior to Echinacea administration, rates of phagocytosis by circulating neutrophils and monocytes were $86 \pm 10\%$, and these data did not change either in control animals (those fed peanut butter alone; $88 \pm 4\%$) or in animals given Echinacea ($92 \pm 8\%$) during the study period.

4. Discussion

Echinacea is an herb historically used to treat a wide variety of ailments ranging from digestive problems to blood infections and snakebites [2], and which has more recently found favor as a popular immune system stimulant with a number of scientific studies suggesting it possesses nonspecific, short-term immune system stimulant properties [1–4]. Data supporting immune stimulation by Echinacea has been provided by a variety in vitro methodologies in which stabilized extracts of the herb have been either administered by injection [10], given as an ethanol extract

[7] or mixed with isolated cells [9]. Since ingestion is the normal mode of administration for human consumption of the drug, none of these methods are likely to reflect true physiological responses as they provide a more rapid introduction of active principles at much higher levels.

In this study, we have shown for the first time that oral ingestion of Echinacea resulted in a more than a 30-fold increase in circulating IL-2 levels—a change that was not observed in any animals administered peanut butter alone (Table 3). Increases in circulating IL-2 occurred only after 3 weeks of oral administration of the herb and were in direct contrast to the differential white blood cell count, which demonstrated an immediate and sustained increase in the percentage of mononuclear cells (neutrophils and lymphocytes) (Fig. 1 and Table 2). Whilst these findings are difficult to rationalize since the source of circulating IL-2 would be the blood lymphocyte population, one putative explanation might be that the active ingredient present in Echinacea that produces these effects may require bioaccumulation in the body prior to providing the IL-2 stimulating effect. In contrast, the stimulus to increase lymphocyte numbers occurred immediately (Table 2), and may either be due to another component of the herbal mixture or require a lower level of Echinacea component in order to produce its effects. Since the Echinacea used in the study is a complex mixture of carbohydrates and lipids, none of which are standardized, any or several of the components could be responsible for the effects seen. However, only animals receiving Echinacea demonstrated these changes and these findings were seen in all the study animals, the results are considered to be reproducible.

Another possible component in the later secretion of IL-2 compared with circulating lymphocyte changes may be the refractoriness of the animals' lymphocytes to external stimulation. Indeed, studies on aging Wistar–Furth [11] and Sprague–Dawley rats

[12] have demonstrated that alterations in circulating lymphocyte number and function are detectable by 13 months of age. In the studies by Kelley et al. [11], decrease in immune function was also reflected in alterations in the capacity of these cells to secrete IL-2 and to react to mitogens such as phytohemagglutinin and concanavalin A.

Previous studies of the effects of *Echinacea* on IL-2 levels in rats have suggested that the herb was without effect on circulating IL-2 levels [11]. These findings may, however, be explained by the use by Kelley et al. [11] of the purified polysaccharide arabinogalactan in place of an *Echinacea* extract. As was previously observed by Wills et al. [4] in a review of the literature, purified polysaccharides from tissue cultures of *E. purpurea* are mainly fucogalactoxyloglucans and arabinogalactans, whereas isolates of aerial components of the plant itself contain the polysaccharide moieties 4-*O*-methyl glucurono-arabinoxylan and arabinorhamnogalactan [16].

Our studies are, however, in agreement with other investigations where aerial components of *Echinacea* were fed to rats and in which no effects were seen on phagocytic activity [8,17]. Studies by Ganguly et al. [13] suggest that with age, rats do show variability in their killing abilities, especially involving macrophages, but that phagocytic ability is relatively normal. Taken together, this would seem to suggest that the lack of responsiveness seen in our study may be more related to the components of *Echinacea* used rather than a true physiological difference. Chemical analyses have determined the presence of two major groups of components in the aerial parts of *Echinacea*; caffeoyl phenols, particularly cichoric acid in *E. purpurea* [16], and alkylamides, which are lipophilic components uncommon in most other species but found in at least two genera of Asteraceae [4]. One explanation for the lack of stimulation of phagocytic activity in the current study is that the components of *Echinacea* previously suggested as effective, namely, alkylamides and polysaccharides are found to be at lower levels in the aerial components of the plant, when compared with the root [4]. The *Echinacea* used in the current study was only standardized for the content of cichoric acid, a caffeic acid derivative [4] with the content of other components being unknown.

Complex mixtures of lipids and carbohydrates, as are found in *Echinacea* extract, are also likely to

provide a strong antigenic impact for the body to which they are introduced. They fulfill the requirements of foreignness, complexity and size necessary for antigenicity [18], and also retain activity even following digestion. Although the exact mode of action of *Echinacea* is still not clearly understood, it is possible that its stimulatory activities allow it to exhibit properties comparable to those of an immunological adjuvant. Widely used to stimulate animal immune systems, these are also complex, metabolically stable, particulate substances normally consisting of complex foreign lipids, carbohydrates and/or proteins, which are able to improve antibody production or vaccine responses [19]. This might explain *Echinacea*'s ability, in at least eight clinical trials, to decrease the severity and reduce the duration of colds and upper respiratory infections, especially when taken early in the course of an infection [1]. Adjuvants are also known to produce intense inflammatory reactions in animals and it is of interest to note that the herb is not recommended for use in patients with autoimmune disease where they may precipitate erythema and edema [20].

The potential of *Echinacea* as a herbal therapeutic is only now being realized and more scientific double-blind studies using standardized over-the-counter preparations of the herb will allow a true evaluation of its activities in human subjects of all ages.

Acknowledgements

This work was supported by a summer research grant given by the Philadelphia University.

References

- [1] Thomas P. *Echinacea*: a natural remedy for the common cold and flu? Nutr Today 2001;36(5):249–53.
- [2] Hobbs C. *Echinacea*: a literature review. HerbalGram 1994;30: 33–48.
- [3] *Echinacea*. The review of natural products. St. Louis: facts and comparisons. Monograph, December 1996.
- [4] Wills RBH, Bone K, Morgan M. Herbal products: active constituents, modes of action and quality control. Nutr Res Rev 2000;13(1):47–77.
- [5] Brevoort P. The booming U.S. botanical market. A new overview. HerbalGram 1998;44:33–48.
- [6] Steinmüller C, Roesler J, Gröttrup E, Franke G, Wagner H,

- Lohmann-Matthes ML. Polysaccharides isolated from plant cell cultures of *Echinacea purpurea* enhance the resistance of immunosuppressed mice against systemic infections with *Candida albicans* and *Listeria monocytogenes*. *Int J Immunopharmacol* 1993;15(5):605–14.
- [7] Bauer VR, Jurcic K, Puhlmann J, Wagner H. Immunologic in vivo and in vitro studies on *Echinacea* extracts. *Arzneimittelforschung* 1988 (Feb);38(2):276–81.
- [8] Schumacher A, Friedberg KD. The effect of *Echinacea angustifolia* on non-specific cellular immunity in the mouse. *Arzneimittelforschung* 1991 (Feb);41(2):141–7.
- [9] Stimpel M, Proksch A, Wagner H, Lohmann-Matthes ML. Macrophage activation and induction of macrophage cytotoxicity by purified polysaccharide fractions from the plant *Echinacea purpurea*. *Infect Immun* 1984;46(3):845–9.
- [10] Luettig B, Steinmüller C, Gifford GE, Wagner H, Lohmann-Matthes ML. Macrophage activation by the polysaccharide arabinogalactan isolated from plant cell cultures of *Echinacea purpurea*. *J Natl Cancer Inst* 1989;81(9):669–75.
- [11] Kelley KW, Brief S, Westly HJ, Novakofski J, Bechtel PJ, Simon J, et al. GH₃ pituitary adenoma cells can reverse thymic aging in rats. *Proc Natl Acad Sci U S A* 1986;83:5663–7.
- [12] Flaherty DK, Wagner CA, Gross CJ, Panyik MA. Aging and lymphocyte subsets in the spleen and peripheral blood of the Sprague–Dawley rat. *Immunopharmacol Immunotoxicol* 1997;19:185–95.
- [13] Ganguly R, Shamblin PB, Craig CP. Microbicidal activity and superoxide production by macrophages in aging rats. *Allerg Immunol* 1984;30:225–9.
- [14] Cundell DR. Generation of histamine and neutrophil chemotactic inhibitory factors by non-typable *Haemophilus influenzae* in vitro. PhD Thesis. London, England: University of London; 1993.
- [15] Hansel FK. Cytologic diagnosis in respiratory allergy and infection. *Ann Allergy* 1966;24:564–9.
- [16] Bauer VR, Jurcic K, Puhlmann J, Wagner H. *Echinacea* species as potential immunostimulatory drugs. In: Wagner H, Farnsworth NR, editors. *Economic and medicinal plant research*, vol. 5. London: Academic Press; 1991. p. 253–321.
- [17] South EH, Exon JH. Multiple immune functions in rats fed *Echinacea* extracts. *Immunopharmacol Immunotoxicol* 2001; 23(3):411–21.
- [18] Janeway CA, Travers P, Walport M, Shlomick M. Appendix 1: immunologists' toolbox: immunization. In: Austin P, Lawrence E, editors. *Immunobiology*. 5th ed. New York: Garland Publishing; 2001. p. 613–5.
- [19] Janeway CA, Travers P, Walport M, Shlomick M. Chapter 14: manipulation of the immune response. In: Austin P, Lawrence E, editors. *Immunobiology*. 5th ed. New York: Garland Publishing; 2001. p. 582–3.
- [20] Ness J, Sherman FT, Pan CX. Alternative medicine: what the data say about common herbal therapies. *Geriatrics* 1999; 54(10):33–6.